

Normal Distribution

Reading Objectives:

In this reading, you will understand the normal and standard normal distributions with some examples.

Main Reading Section:

Definition 1: A random variable is said to be **normally distributed** with parameters μ and σ^2 , and we write $X \sim N(\mu, \sigma^2)$ if its density is

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/2\sigma^2}, -\infty < x < \infty$$

The normal density $f(x)$ is a bell-shaped curve that is symmetric about μ and attains its maximum value of $\frac{1}{\sqrt{2\pi}\sigma} \approx 0.399/\sigma$ at $x = \mu$.

Definition 2: It follows from the foregoing that if $X \sim N(\mu, \sigma^2)$, then $Z = \frac{X-\mu}{\sigma}$ is a normal random variable with mean 0 and variance 1. Such a random variable Z is said to have a standard, or unit, normal distribution.

Let $\Phi(\cdot)$ denote its distribution function. That is

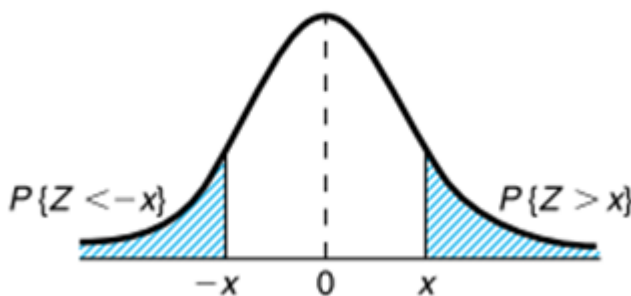
$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-y^2/2} dy, -\infty < x < \infty$$

This result that $Z = (X - \mu)/\sigma$ has a standard normal distribution when X is normal with parameters μ and σ^2 is quite important, for it enables us to write all probability statements about X in terms of probabilities for Z .

Note: For instance, to obtain $P\{X < b\}$, we note that X will be less than b if and only if $\frac{X-\mu}{\sigma}$ is less than $\frac{b-\mu}{\sigma}$, and so

$$P\{X < b\} = P\left\{\frac{X-\mu}{\sigma} < \frac{b-\mu}{\sigma}\right\} = \Phi\left(\frac{b-\mu}{\sigma}\right)$$

It remains for us to compute $\Phi(x)$



We can also obtain $\Phi(-x)$ from the table using the symmetry (about 0) of the standard normal probability density function. That is, for $x > 0$, if Z represents a standard normal random variable, then

$$\begin{aligned}\Phi(-x) &= P\{Z < -x\} \\ &= P\{Z > x\} = 1 - \Phi(x)\end{aligned}$$

Example 1. If X is a normal random variable with mean $\mu = 3$ and variance $\sigma^2 = 16$, find

- a) $P\{X < 11\}$
- b) $P\{X > -1\}$
- c) $P\{2 < X < 7\}$

Solution:

- a) $P\{X < 11\} = P\left\{\frac{X-3}{4} < \frac{11-3}{4}\right\} = \Phi(2) = 0.9772$
- b) $P\{X > -1\} = P\left\{\frac{X-3}{4} > \frac{-1-3}{4}\right\} = P\{Z > -1\} = P\{Z < 1\} = 0.8413$
- c) $P\{2 < X < 7\} = P\left\{\frac{2-3}{4} < \frac{X-3}{4} < \frac{7-3}{4}\right\} = \Phi(1) - \Phi(-1/4) = \Phi(1) - (1 - \Phi(1/4))$
 $= 0.8413 + 0.5987 - 1 = 0.4400$

Example 2: Students of a class were given an aptitude test. Their marks were found to be normally distributed, with mean 60 and a standard deviation 5. What percentage of students scored more than 60 marks?

Solution: $X = 60$, $\mu = 60$, and $\sigma = 5$

$$Z = \frac{X - \mu}{\sigma} = \frac{60 - 60}{5} = 0$$

If $x > 60$, then $Z > 0$

The area lying to the right of z is 0.5.

The percentage of students getting more than 6 marks is 50%.

Example 3: The mean inside diameter of a sample of 200 washers produced by a machine is 0.502, and the standard deviation is 0.005 cm. The purpose for which these washers are intended is to allow a maximum tolerance in the diameter of 0.496 to 0.508 cm; otherwise, the washers are considered defective. Determine the percentage of defective washers the machine produces, assuming the diameters are normally distributed.

Solution: $Z_1 = \frac{X - \mu}{\sigma} = \frac{0.496 - 0.502}{0.005} = -1.2$

$$Z_2 = \frac{X - \mu}{\sigma} = \frac{0.508 - 0.502}{0.005} = 1.2$$

Area for non-defective washers = Area between $Z = -1.2$ and $Z = 1.2$

= 2. Area between $Z=0$ and $Z = 1.2$

= $2(0.3849) = 0.7698 = 76.98\%$

Percentage of defective washers = $100 - 76.98 = 23.02\%$